

JAE HONG LEE

585-957-8491 • ll.leejaehong.ll@gmail.com • linkedin.com/in/hong-lee-0821/ • github.com/digitaldna01 • jhonglee.com

EDUCATION

Boston University, Boston, MA

B.A., Computer Science / Minor in Visual Arts

Computing and Technical Honor Society / Upsilon Pi Epsilon (UPE)

May 2025

3.90 GPA | Magna Cum Laude

PROFESSIONAL EXPERIENCE

Agentic AI Software Engineer

LG CNS

Jan 2026 – Current

Seoul, Korea

- Built a Retrieval-Augmented Generation (RAG) system for MES manuals by designing preprocessing pipelines (PDF parsing, semantic chunking, metadata injection) and storing embeddings in pgvector.
- Implemented LLM-based query processing (rewriting, hybrid retrieval with BM25 + vector similarity, reranking) within a modular FastAPI backend, ensuring accurate responses, extensibility, and enterprise integration.

Machine Learning Specialist

LSAT Demon LLC

Jun 2025 – Aug 2025

Remote

- Developed an LLM-powered backend system for personalized text generation, integrating FastAPI pipelines with semantic retrieval (BM25, FAISS) to enhance accuracy and reliability.
- Curated domain datasets and applied advanced prompt engineering strategies, enabling scalable deployment with modern LLMs (e.g., Mistral, GPT).

System Optimization Research Intern

Korea Electronics Technology Institute

Jun 2024 – Aug 2024

Seungnam, Korea

- Contributed to research optimizing memory efficiency for quantum simulations by storing only measured qubits up to 45 qubits, and developed a quantum circuit simulator using Google's Tensor Network, qiskit, and statevector to perform Grover's algorithm.
- Acquired comprehensive knowledge in quantum computing through foundational studies and delivered two internal seminars on key concepts such as qubits, superposition, Bell's theorem, the EPR paradox, and quantum computing simulations.

Computer Vision Research Intern

Korea Electronics Technology Institute

Jun 2022 – Aug 2022

Seungnam, Korea

- Applied Google's Vision Transformer (ViT) for advanced breed classification in cats and dogs kaggle dataset, incorporating patch based Transformer and Multi-head Attention mechanisms, achieving a classification accuracy of over 95% on benchmark datasets.
- Re-engineered the ViT model using TensorFlow, developing an API and integrating it with an open CV camera module for real-time capture applications, enabling accurate breed classification for real-life pets, improving both speed and precision.

RESEARCH PROJECTS

Text-to-Panorama Generation Undergraduate Research

Research Assistant, Advisor: Aoming Liu

April 2024 - Present

Boston, MA

- Investigating a cube-based approach to refine text-to-360-degree panorama generation by combining spot diffusion techniques with latent space noise projection, with the goal of creating depth and improving visual continuity.
- Focused on achieving seamless transitions and realism in panorama generation by completing a 3D camera tutorial, covering areas of camera calibration, object localization, and geometric distortion correction to improve accuracy and depth perception.

AI-driven Hand Pose estimation Undergraduate Research

Research Assistant, Advisor: Eung-Joo Lee, Jong In Choi

Sep 2023 - May 2024

- Implemented a custom-built haptic controller to generate natural hand postures, trained the XGBoost model with a collection of 3000-9000 training datasets, which improved handpose accuracy as dataset size increased.
- Designed a technique for generating hand poses and identified the need for time-series methods to reduce tremors in haptic control.

SOFTWARE ENGINEER PROJECTS

To-Do Calendar

React.js, Google Calender & Authentication API, Firebase, CSS, Figma

Jan 2024 – May, 2024

Boston, MA

- Integrated task management with Google Calendar to sync tasks and deadlines, improving productivity by providing an overview.
- Automatic meeting additions have been implemented for Google Meet and Zoom to ensure users won't miss a meeting and to visually link tasks to their respective deadlines.

PUBLICATIONS

Jongin Choi, Jaehong Lee, Daniel Oh, and Eung-Joo Lee, "Machine Learning-Based Hand Pose Generation Using a Haptic Controller."Electronics 13.10 (2024): 1970